

RAAKESH MENON A

Senior Data Engineer

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SUMMARY

Senior Data Engineer with 4+ years delivering enterprise data platforms for Fortune 500 global clients across banking, semiconductor manufacturing, aerospace manufacturing, and animal-health domains. Specializes in Snowflake data warehousing, distributed processing with Apache Spark/PySpark, and Python-driven ETL/ELT modernization — re-architecting legacy pipelines into scalable cloud-native frameworks with automated data-quality, validation, and anomaly-detection controls. Hands-on with Azure DevOps CI/CD and building on the Azure data engineering stack (ADLS Gen2, ADF, Azure Databricks, Delta Lake); owns Agile delivery end-to-end from requirements through production support.

CORE TECHNICAL SKILLS

Programming: Python (Pandas, NumPy), Advanced SQL, PL/SQL, Shell Scripting, REST API Integration

Data Engineering: Snowflake, Apache Spark, PySpark, ETL/ELT Development & Modernization, Data Pipeline Architecture, Apache Airflow, Apache Kafka (Streaming Ingestion), Talend, Workflow Orchestration, Incremental Loads & CDC, Data Migration, Data Warehousing, Dimensional Data Modelling (Star Schema), Data Integration

Azure Data Platform: Azure Data Lake Storage (ADLS Gen2), Azure Blob Storage, Azure Data Factory (ADF), Azure Databricks, Delta Lake, Azure Synapse Analytics, Azure SQL Database, Azure Functions, Azure Key Vault, Azure Monitor, Microsoft Fabric, Microsoft Purview

Architecture & Governance: Data Lake / Lakehouse, Medallion Architecture, Data Quality Engineering, Data Validation Frameworks, Anomaly Detection, Data Profiling, Data Lineage, Data Governance, Query & Performance Optimization

Machine Learning & AI: Anomaly Detection, Time-Series Forecasting, Regression, Random Forest, Gradient Boosting (XGBoost), Feature Engineering, Model Evaluation, MLflow (Experiment Tracking & Model Registry), MLOps, Generative AI, LLM Integration, Prompt Engineering, Retrieval-Augmented Generation (RAG), LLM-Assisted Data Engineering

BI & Analytics: Power BI, DAX, Power Query, Tableau, Looker Studio, KPI Dashboard Development

DevOps & Delivery: Azure DevOps (Boards, Repos, Pipelines), Git, CI/CD, Agile/Scrum, Jira, SDLC, Stakeholder Management

PROFESSIONAL EXPERIENCE

Senior Systems Engineer (Senior Data Engineer)

Apr 2023 – Present

Tata Consultancy Services (TCS) — Bengaluru, India

- Designed and optimized Apache Spark and PySpark pipelines processing **~2 TB per month** of structured and semi-structured data for a global banking client, cutting distributed-processing bottlenecks by ~35% through **partition pruning and broadcast-join tuning**.
- Architected Snowflake-based data integration and Power BI analytics for a leading semiconductor manufacturer, unifying BOM hierarchies, SAP, and Teamcenter engineering data from **8** source systems and cutting dashboard refresh turnaround by ~30% — strengthening visibility into manufacturing KPIs for engineering and business stakeholders.
- Built Python automation frameworks for data validation, reconciliation, anomaly detection, and statistical trend analysis across **40+** datasets, cutting manual QA effort by ~50% and improving data-quality visibility.
- Automated enterprise reporting workflows in Power BI, reducing end-to-end reporting cycle time by ~75% (**from ~2 days to ~4 hours**) and improving operational decision-making across banking platforms.
- Led migration of **25+** legacy **Oracle/on-prem** workloads to Snowflake for banking platforms, redesigning warehouse and data models with **zero-downtime cutover**.
- Directed Agile delivery through Azure DevOps (Boards, Repos, Pipelines), owning CI/CD execution and release management for reporting modernization with consistently on-time sprint releases and minimal production defects.

Systems Engineer (Data Engineer)

Dec 2021 – Mar 2023

Tata Consultancy Services (TCS) — Bengaluru, India

- Re-architected **30+** legacy Perl-based ETL workflows into Python-driven pipelines and integrated Talend orchestration for enterprise banking platforms, cutting dependency-management overhead by ~40% and orchestration overhead by ~25%.
- Developed scalable Python and SQL ETL workflows for high-volume aerospace manufacturing datasets, improving analytics processing performance by ~30% and enabling business-intelligence modernization for manufacturing operations.
- Built Snowflake and Python ETL pipelines with workflow orchestration for enterprise animal-health datasets, improving data-integration reliability across **6** source systems.
- Delivered 15+ reusable ETL components and automated validation/transformation frameworks adopted across **5+ project teams**, cutting manual processing effort by ~40% and accelerating reporting delivery timelines.

KEY PROJECTS

Five end-to-end data platforms (portfolio builds; results measured on synthetic datasets — full code and pipelines on GitHub).

LLM-Augmented Legacy ETL Modernization Framework — *Python, Claude API, sqlglot, PySpark, Great Expectations, pytest, Airflow, Snowflake, Azure DevOps*

- Built a framework that translates legacy Perl/SQL ETL to **Python** using an **LLM**, then proves behavioural equivalence by dual-executing legacy and translated jobs on the same data and diffing outputs cell-by-cell — a 5-component weighted equivalence score (schema, row set, cell values, null patterns, ordering) with a **CI** gate blocking promotion below 99%.
- Achieved 85.7% auto-translation with 99.63% mean equivalence and 0 defects past the gate on a 7-unit corpus; auto-generated 129 regression tests pinned to legacy behaviour; sandboxed **LLM**-generated code via AST-based import allow-listing as a **CI** supply-chain security control.

Aerospace MRO Supply-Chain Digital Twin — *Python, PySpark, XGBoost/scikit-learn, Snowflake, Airflow, Power BI*

- Unified 5 heterogeneous sources (SAP extracts, OAuth2/EDI supplier portals, customs XML, WMS) into a **Snowflake** star-schema hub with **SCD Type 2** dimensions, deterministic SHA-256 surrogate keys, idempotent MERGE loads, and fail-soft ingestion; **PySpark** part-identity resolution across supersession chains with skew-aware salted joins.
- Engineered intermittent-demand forecasting (Poisson-objective **XGBoost**, Syntetos-Boylan classification, time-based splits) evaluated on MASE 0.86 vs a seasonal-naive baseline; flagged 24 parts pre-stockout with a median 30-day warning, with an **Airflow** gate refusing to publish any model at MASE ≥ 1.0 .

Semiconductor Yield Intelligence Pipeline — *Python, networkx, scikit-learn, Snowflake (recursive CTEs), dbt, Airflow, Power BI*

- Built BOM-version-aware root-cause analysis for wafer-yield excursions: 8,000-node BOM explosion in 0.13s (implemented twice — **Python** graph engine and **Snowflake** recursive CTE for BI/**dbt**), per-lot as-built feature matrices, and a random-forest ranker using permutation importance, empirical-Bayes shrinkage, and robust MAD excursion thresholds.
- Recovered 3 of 3 injected root causes ranked #1/#2/#3 in 34 seconds (vs ~2–3 days manual analysis) at holdout R^2 0.885, enforced by a **CI** ground-truth gate that fails the build if the pipeline stops recovering known causes.

FHIR-Native Clinical Data Lakehouse with Privacy-Preserving Record Linkage — *Python, HL7apy, FHIR R4, Delta Lake (SCD2), PySpark, Great Expectations, Airflow*

- Implemented Bloom-filter PPRL (2048-bit composite CLKs, HMAC-SHA256 with field-salted keys, Kirsch-Mitzenmacher double hashing) so patient matching across two hospital systems never transmits raw PII; Fellegi-Sunter match/review/non-match thresholds with union-find golden-record clustering and deterministic enterprise patient IDs.
- Achieved 100% match precision (zero false merges) and 97.5% recall, cutting duplicate rate from 8.0% to 0.2%, with per-facility data-quality scoring (completeness, temporal validity, uniqueness) on an **HL7v2** + **FHIR R4** ingestion Lakehouse.

Real-Time Banking Regulatory Reporting & Reconciliation Platform — *PySpark on Azure Databricks, Delta Lake, ADLS Gen2 (medallion), Azure Event Hubs, Airflow, Power BI, Azure DevOps*

- Built a Bronze/Silver/Gold medallion Lakehouse reconciling **SWIFT MT940**, general-ledger, and core-banking feeds at position level with automated break classification (matched / amount-break / missing-in-source) and per-source data-quality rule engines.
- Automated **Basel III** regulatory computation (CAR, LCR, SLR vs RBI minimums, risk-weighted assets) on **Databricks SQL** + **Power BI**, cutting the reconciliation cycle from T+1 to under 30 minutes with ~98% automated break detection.

ACHIEVEMENTS

- Recognized by client stakeholders and internal leadership with **multiple client appreciation awards** for delivering 4+ high-impact data engineering and analytics solutions improving operational efficiency and reporting reliability.
- Conducted **10+** technical knowledge-sharing and upskilling sessions on Python, SQL, and ETL best practices for **25+** engineers, strengthening team capability in analytics engineering.

EDUCATION

Bachelor of Engineering, Mechanical Engineering — Anna University, Chennai, India

2016 – 2020

CERTIFICATIONS

- Google Advanced Data Analytics Professional Certificate
- IBM Data Science Professional Certificate
- Google Business Intelligence Professional Certificate
- Generative AI with Large Language Models — DeepLearning.AI
- **[Add on completion: Microsoft DP-700 / Azure Data Engineer Associate; Databricks Data Engineer Associate]**